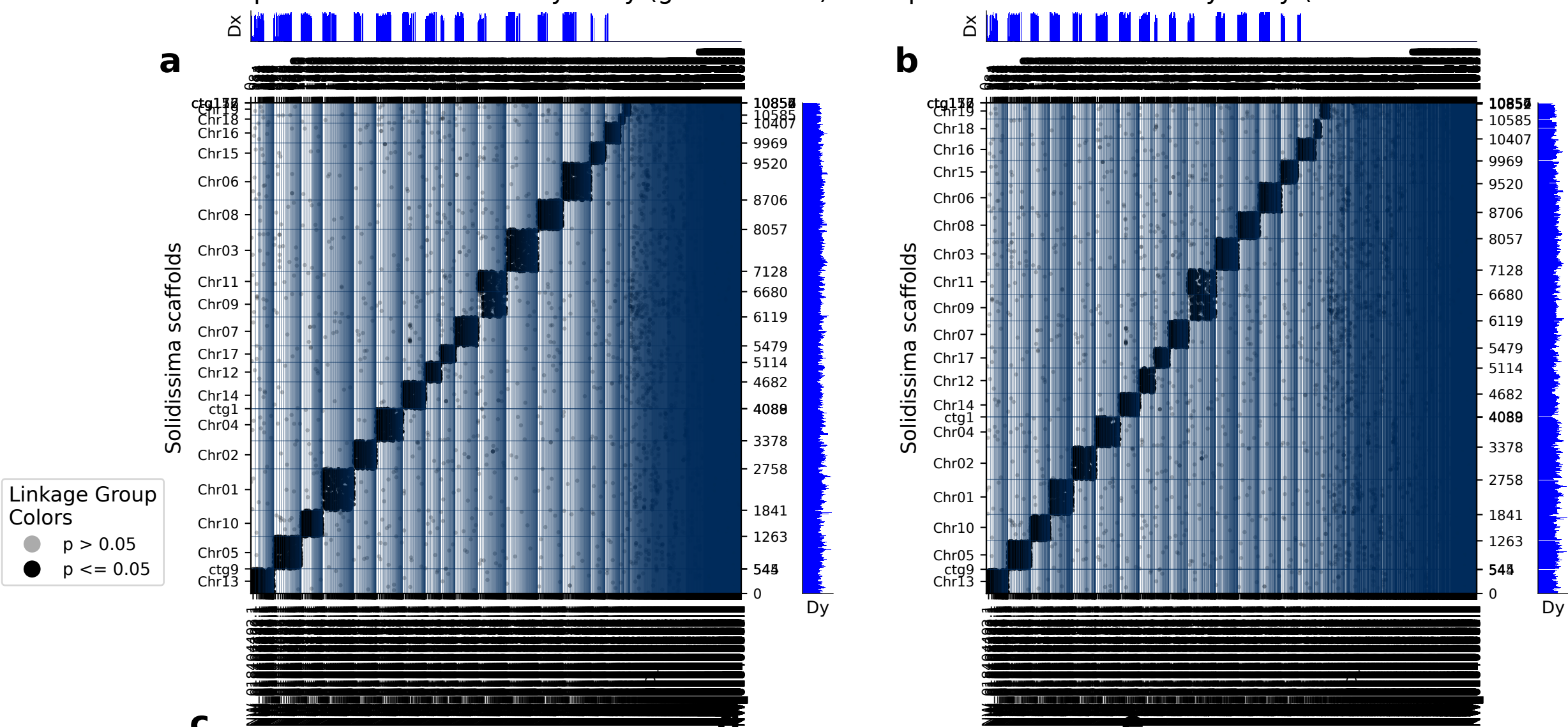


diamond: 10858 orthologs on 1556 Mizuhopecten scaffolds and 24 Solidissima scaffolds.

Mizuhopecten vs Solidissima synteny (gene indices)Mizuhopecten vs Solidissima synteny (chromosome coordinates)



c Mizuhopecten vs Solidissima scaffolds **d** Mizuhopecten vs Solidissima FET, whole chromosome **e** Mizuhopecten vs Solidissima FET, chrom. pieces

	Mizuhopecten scaf	Solidissima scaf	p (FET)	Count
0	NW_018404492.1	Chr13	6.18e-87	82
1	NW_018408896.1	Chr10	3.59e-68	60
2	NW_018408902.1	Chr05	7.03e-57	58
3	NW_018405386.1	Chr01	1.08e-54	58
4	NW_018407390.1	Chr13	5.70e-57	55
5	NW_018404940.1	Chr02	2.32e-57	52
6	NW_018480077.1	Chr05	5.45e-46	50
7	NW_018406397.1	Chr04	1.23e-50	48
8	NW_018407065.1	Chr05	2.67e-43	43
9	NW_018404097.1	Chr10	6.38e-45	41
10	NW_018408465.1	Chr14	1.43e-39	40
11	NW_018407700.1	Chr04	2.22e-36	40
12	NW_018408713.1	Chr10	2.23e-42	39
13	NW_018407435.1	Chr10	2.23e-42	39
14	NW_018408653.1	Chr08	3.78e-38	38
15	NW_018408629.1	Chr07	4.10e-35	38
16	NW_018405944.1	Chr03	1.25e-29	38
17	NW_018408899.1	Chr06	2.60e-35	37
18	NW_018404662.1	Chr05	1.51e-29	37
19	NW_018408761.1	Chr07	5.77e-36	36
20	NW_018408252.1	Chr12	3.17e-35	35
21	NW_018407509.1	Chr01	3.64e-30	35
22	NW_018404373.1	Chr06	7.85e-36	35
23	NW_018403543.1	Chr06	4.83e-31	35
24	NW_018408638.1	Chr03	1.06e-32	34
25	NW_018480076.1	Chr06	4.77e-32	33
26	NW_018404269.1	Chr11	9.99e-37	33
27	NW_018480078.1	Chr08	4.15e-30	32
28	NW_018408887.1	Chr03	4.70e-29	32
29	NW_018408880.1	Chr17	1.07e-32	32
30	NW_018406415.1	Chr10	6.29e-29	32
31	NW_018404481.1	Chr14	6.55e-37	32
32	NW_018408742.1	Chr12	2.71e-36	31
33	NW_018408154.1	Chr07	5.60e-27	31
34	NW_018408916.1	Chr11	8.20e-27	30
35	NW_018408895.1	Chr03	9.44e-26	30
36	NW_018408645.1	Chr06	1.71e-27	30
37	NW_018406089.1	Chr05	3.09e-27	30
38	NW_018408921.1	Chr12	2.36e-30	29
39	NW_018408919.1	Chr02	7.29e-30	29
40	NW_018406060.1	Chr04	3.98e-27	29
41	NW_018405900.1	Chr14	1.33e-31	29
42	NW_018407364.1	Chr02	1.25e-28	28
43	NW_018405342.1	Chr14	3.50e-29	28
44	NW_018403589.1	Chr12	3.05e-31	28
45	NW_018408914.1	Chr04	8.21e-25	27
46	NW_018408913.1	Chr03	3.91e-25	27
47	NW_018408770.1	Chr06	1.04e-26	27
48	NW_018406690.1	Chr06	1.46e-21	27
49	NW_018405747.1	Chr05	1.11e-24	27
50	NW_018404519.1	Chr07	1.40e-29	27
51	NW_018408726.1	Chr15	6.40e-29	26
52	NW_018408623.1	Chr08	9.13e-27	26
53	NW_018407907.1	Chr05	1.73e-24	26
54	NW_018407426.1	Chr12	1.82e-31	26
55	NW_018406268.1	Chr01	8.27e-23	26
56	NW_018405113.1	Chr04	5.25e-19	26
57	NW_018404308.1	Chr05	1.32e-25	26
58	NW_018407733.1	Chr06	5.98e-22	25
59	NW_018407511.1	Chr03	5.62e-23	25
60	NW_018406874.1	Chr13	2.30e-26	25
61	NW_018408820.1	Chr16	1.23e-25	24
62	NW_018408786.1	Chr09	1.35e-21	24
63	NW_018407102.1	Chr01	1.12e-18	24

	Mizuhopecten scaf	Solidissima scaf	p (FET)	Count
0	NW_018404492.1 : 0-end	Chr13 : 0-end	6.18e-87	82
1	NW_018408896.1 : 0-end	Chr10 : 0-end	3.59e-68	60
2	NW_018408902.1 : 0-end	Chr05 : 0-end	7.03e-57	58
3	NW_018405386.1 : 0-end	Chr01 : 0-end	1.08e-54	58
4	NW_018407390.1 : 0-end	Chr13 : 0-end	5.70e-57	55
5	NW_018404940.1 : 0-end	Chr02 : 0-end	2.32e-57	52
6	NW_018480077.1 : 0-end	Chr05 : 0-end	5.45e-46	50
7	NW_018406397.1 : 0-end	Chr04 : 0-end	1.23e-50	48
8	NW_018407065.1 : 0-end	Chr05 : 0-end	2.67e-43	43
9	NW_018404097.1 : 0-end	Chr10 : 0-end	6.38e-45	41
10	NW_018408465.1 : 0-end	Chr14 : 0-end	1.43e-39	40
11	NW_018407700.1 : 0-end	Chr04 : 0-end	2.22e-36	40
12	NW_018408713.1 : 0-end	Chr10 : 0-end	2.23e-42	39
13	NW_018407435.1 : 0-end	Chr10 : 0-end	2.23e-42	39
14	NW_018408653.1 : 0-end	Chr08 : 0-end	3.78e-38	38
15	NW_018408629.1 : 0-end	Chr07 : 0-end	4.10e-35	38
16	NW_018405944.1 : 0-end	Chr03 : 0-end	1.25e-29	38
17	NW_018408899.1 : 0-end	Chr06 : 0-end	2.60e-35	37
18	NW_018404662.1 : 0-end	Chr05 : 0-end	1.51e-29	37
19	NW_018408761.1 : 0-end	Chr07 : 0-end	5.77e-36	36
20	NW_018408252.1 : 0-end	Chr12 : 0-end	3.17e-35	35
21	NW_018407509.1 : 0-end	Chr01 : 0-end	3.64e-30	35
22	NW_018404373.1 : 0-end	Chr06 : 0-end	7.85e-36	35
23	NW_018403543.1 : 0-end	Chr06 : 0-end	4.83e-31	35
24	NW_018408638.1 : 0-end	Chr03 : 0-end	1.06e-32	34
25	NW_018480076.1 : 0-end	Chr06 : 0-end	4.77e-32	33
26	NW_018404269.1 : 0-end	Chr11 : 0-end	9.99e-37	33
27	NW_018480078.1 : 0-end	Chr08 : 0-end	4.15e-30	32
28	NW_018408887.1 : 0-end	Chr03 : 0-end	4.70e-29	32
29	NW_018408880.1 : 0-end	Chr17 : 0-end	1.07e-32	32
30	NW_018406415.1 : 0-end	Chr10 : 0-end	6.29e-29	32
31	NW_018404481.1 : 0-end	Chr14 : 0-end	6.55e-37	32
32	NW_018408742.1 : 0-end	Chr12 : 0-end	2.71e-36	31
33	NW_018408154.1 : 0-end	Chr07 : 0-end	5.60e-27	31
34	NW_018408916.1 : 0-end	Chr11 : 0-end	8.20e-27	30
35	NW_018408895.1 : 0-end	Chr03 : 0-end	9.44e-26	30
36	NW_018408645.1 : 0-end	Chr06 : 0-end	1.71e-27	30
37	NW_018406089.1 : 0-end	Chr05 : 0-end	3.09e-27	30
38	NW_018408921.1 : 0-end	Chr12 : 0-end	2.36e-30	29
39	NW_018408919.1 : 0-end	Chr02 : 0-end	7.29e-30	29
40	NW_018406060.1 : 0-end	Chr04 : 0-end	3.98e-27	29
41	NW_018405900.1 : 0-end	Chr14 : 0-end	1.33e-31	29
42	NW_018407364.1 : 0-end	Chr02 : 0-end	1.25e-28	28
43	NW_018405342.1 : 0-end	Chr14 : 0-end	3.50e-29	28
44	NW_018403589.1 : 0-end	Chr12 : 0-end	3.05e-31	28
45	NW_018408914.1 : 0-end	Chr04 : 0-end	8.21e-25	27
46	NW_018408913.1 : 0-end	Chr03 : 0-end	3.91e-25	27
47	NW_018408770.1 : 0-end	Chr06 : 0-end	1.04e-26	27
48	NW_018406690.1 : 0-end	Chr06 : 0-end	1.46e-21	27
49	NW_018405747.1 : 0-end	Chr05 : 0-end	1.11e-24	27
50	NW_018404519.1 : 0-end	Chr07 : 0-end	1.40e-29	27
51	NW_018408726.1 : 0-end	Chr15 : 0-end	6.40e-29	26
52	NW_018408623.1 : 0-end	Chr08 : 0-end	9.13e-27	26
53	NW_018407907.1 : 0-end	Chr05 : 0-end	1.73e-24	26
54	NW_018407426.1 : 0-end	Chr12 : 0-end	1.82e-31	26
55	NW_018406268.1 : 0-end	Chr01 : 0-end	8.27e-23	26
56	NW_018405113.1 : 0-end	Chr04 : 0-end	5.25e-19	26
57	NW_018404308.1 : 0-end	Chr05 : 0-end	1.32e-25	26
58	NW_018407733.1 : 0-end	Chr06 : 0-end	5.98e-22	25
59	NW_018407511.1 : 0-end	Chr03 : 0-end	5.62e-23	25
60	NW_018406874.1 : 0-end	Chr13 : 0-end	2.30e-26	25
61	NW_018408820.1 : 0-end	Chr16 : 0-end	1.23e-25	24
62	NW_018408786.1 : 0-end	Chr09 : 0-end	1.35e-21	24
63	NW_018407102.1 : 0-end	Chr01 : 0-end	1.12e-18	24

(a) depicts the chromosomes/scaffolds of Mizuhopecten (x-axis) plotted against the chromosomes/scaffolds of Solidissima (y-axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10858 orthologs found between 1556 Mizuhopecten scaffolds and 24 Solidissima scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes. (c) shows where gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as panel c, but does not show the gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.